

Calendar

REAL VARIABLES I

Fall 2026

I will put all deadlines on your [Blackboard Calendar](#). Please check it regularly!

0.1 Key university dates

For official University dates, see the [Fall 2026 academic calendar](#) and the [Registrar's final-exam schedule](#).

Date	Event
Monday, August 17	First day of classes
Friday, August 21	Last day to add this course through the standard add period
Friday, August 28	Last day to drop without a W
Thursday, December 3	Last day of classes and last day to withdraw from all classes

0.2 Exams

Date	Event
Week of Wednesday, October 12 - Oct 18, 2026, TENTATIVE	Written midterm (two hours, outside the regular class)
Wednesday, December 9, 2026, 10:15 - 12:15	Final-exam period

0.3 Daily lecture log

	Wk	Lec
		L1: Riemann integration and its limitations
2026-08-17		Review: the Riemann integral; theorem+proof: continuous functions are Riemann integrable
		L2: Countability and unbounded functions
2026-08-19		Proposition + proof: continuous functions with L1 norm are not a complete metric space. L
		L3: Countability, the extended reals, and sums of non-negative numbers
2026-08-21		Review: fat rationals; sketch: Riemann integrable functions with respect to L^1 are incomplete
		L4: Sums of non-negative numbers
2026-08-22		Review: continuity of arbitrary non-negative sums with respect to increasing index sets Cor
		L5: Outer measures generated by covering costs
2026-08-26		Motivation: construct a notion of size from elementary sets whose costs are already known.
		L6: Exercise session: Monotone and dominated convergence for sums
2026-08-28		Exercise session: Monotone and dominated convergence for sums
		L7: Lebesgue outer measure on $\mathbb{R}^d$
2026-08-30		Prop+proof: maximality of generated outer measures. Defn: axis-parallel rectangles, volume
		L8: Lebesgue outer measure and the Vitali obstruction
2026-09-02		Prop+proof: translation invariance and dilation scaling of λ^* . Defn: Vitali equivalence relation
		L9: Exercise session: HW02
2026-09-04		Conclusion of proof: rational balls as generators of Borels on $\mathbb{R}^d$ Exercise session: HW02.
		No class: Labor Day
2026-09-07		Labor Day holiday; no class.
		L10: Measures and their basic inequalities
2026-09-09	TENTATIVE	Defn: measures and measure spaces. Prop+proof: finite additivity, monotonicity, and the d
		L11: Carathéodory measurable sets, σ-algebras and measures: examples
2026-09-11		Thm+proof(conclusion): Carathéodory measurable sets are a σ algebra and μ^* is a measure
		L12: Dynkin systems and measurable functions
2026-09-14		Defns: finite, probability, and σ -finite measures. Defns: π -systems and Dynkin (λ -)systems.
		L13:
2026-09-16	TENTATIVE	
		L14: Carathéodory measurability II: the extension theorem
2026-09-18	TENTATIVE	Thm+proof: Carathéodory measurable sets form a σ -algebra. Thm+proof: restriction of μ^*
		L15: Lebesgue measurable sets and Lebesgue measure
2026-09-21	TENTATIVE	Prop+proof: Carathéodory measurability of half-spaces and rectangles. Cor: open and Borel
		L16: Dynkin's π-λ theorem and uniqueness of measures
2026-09-23	TENTATIVE	Defns: π -systems and Dynkin (λ -)systems. Thm+proof: Dynkin's π - λ theorem. Thm+proof
		L17: Measurable functions
2026-09-25	TENTATIVE	Defn: measurable maps and pullback σ -algebras. Prop: inverse-image operations and comp
	TENTATIVE	Future lecture topics and their placement may change as the course develops.